

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2457166.7514 +/- 0.00077 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.17 +/- 0.01
Transit depth (Rp/Rs)^2	2.88 +/- 0.35 [%]
Orbital Inclination (inc)	85.53 +/- 0.76
Airmass coefficient 1 (a1)	1.292 +/- 0.021
Airmass coefficient 2 (a2)	-0.1904 +/- 0.0041
Scatter in the residuals of the lightcurve fit is	1.6 %
Best Comparison Star	None
Optimal Aperture	2.79
Optimal Annulus	9.99
Transit Duration (day)	0.0876 +/- 0.0081

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.17 $\pm$ 0.01 vs 0.1394 (deviation 2.39 σ)
Duration fit vs published	0.0876 d vs 0.0983 d (deviation 1.03 σ)
Mid-time O-C	2.6 min
Depth SNR	13.3
Residual scatter	1.6 %

Light curve

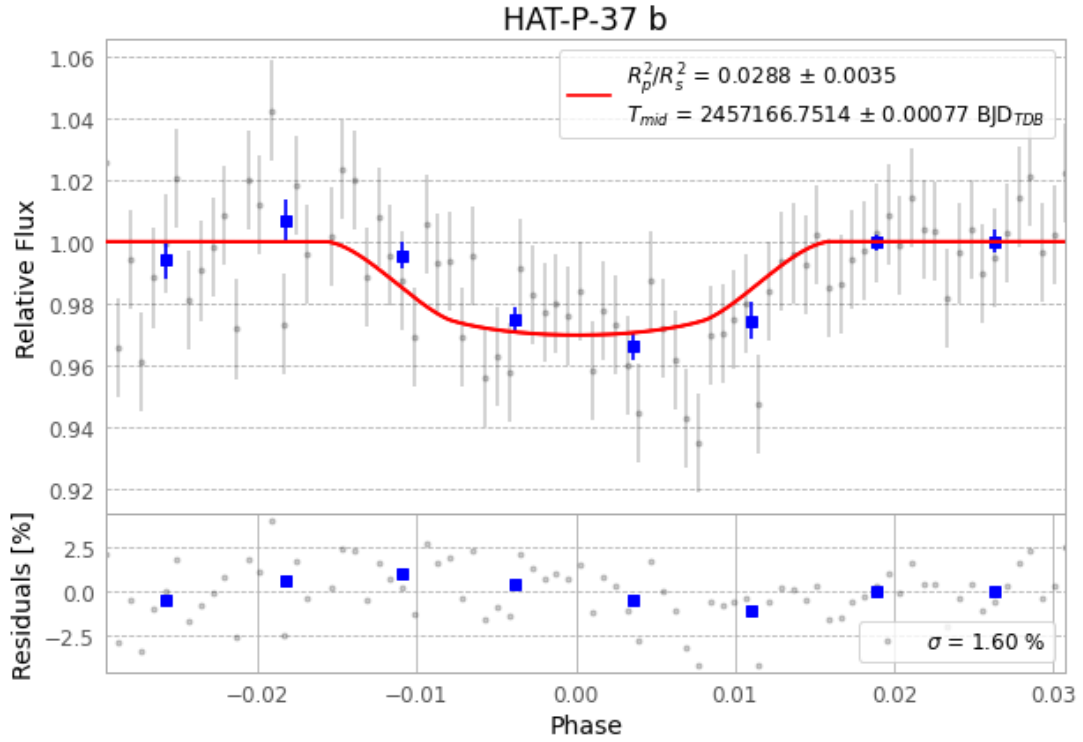


Figure 1 — FinalLightCurve_HAT-P-37 b_2015-05-23.png (SHA-256 e81c6ddd3c12da64...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [230, 233], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e9a011cfe08cf1d4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

82 FITS frames (54 MB), HATP-37150524040012.FITS → HATP-37150524080312.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-150524.log (97fc8ac96885...), FinalLightCurve_HAT-P-37 b_2015-05-23.png (e81c6ddd3c12...), FinalLightCurve_HAT-P-37 b_2015-05-23.csv (11b47ebb349c...)

Compute resources

Wall time 123.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-24 01:28Z.